

Application

Example

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

The transformation T takes a vector $\mathbf{v} \in \mathbb{R}^2$ and maps it to $A\mathbf{v}$.

Step 1: Verify if T is One-to-One

To determine if T is one-to-one, we need to check whether the matrix A is invertible. If A is invertible, then T is one-to-one.

The invertibility of a matrix can be checked by calculating its **determinant**. If the determinant is non-zero, the matrix is invertible, and the transformation is one-to-one.

The determinant of A is:

$$\det(A) = (2)(3) - (1)(1) = 6 - 1 = 5$$

Since $\det(A) = 5 \neq 0$, the matrix A is invertible, and therefore the transformation T is one-to-one.

Step 2: Apply the Linear Transformation

Let's apply the transformation T to some vectors to see how it behaves.

- Take $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Applying A to $\mathbf{v}_1$:

$$T(\mathbf{v}_1) = A\mathbf{v}_1 = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\text{So, } T(\mathbf{v}_1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

- Take $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Applying A to $\mathbf{v}_2$:

$$T(\mathbf{v}_2) = A\mathbf{v}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\text{So, } T(\mathbf{v}_2) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Step 3: Check the Uniqueness of Mappings

We observe that:

$$T(\mathbf{v}_1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \text{and} \quad T(\mathbf{v}_2) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Since $T(\mathbf{v}_1) \neq T(\mathbf{v}_2)$, the transformation maps distinct vectors to distinct vectors, reinforcing the fact that T is one-to-one.

Step 4: Solve the Equation $T(\mathbf{v}) = 0$

To further verify that the transformation is one-to-one, we can solve $T(\mathbf{v}) = A\mathbf{v} = 0$ for $\mathbf{v}$.

This requires solving the system:

$$\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This corresponds to the system of equations:

$$2v_1 + v_2 = 0$$

$$v_1 + 3v_2 = 0$$

From the first equation, we have:

$$v_2 = -2v_1$$

Continue..

Substituting this into the second equation:

$$v_1 + 3(-2v_1) = 0$$

$$v_1 - 6v_1 = 0$$

$$-5v_1 = 0$$

$$v_1 = 0$$

Thus, $v_1 = 0$ and from $v_2 = -2v_1$, we get $v_2 = 0$. Therefore, $\mathbf{v} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Since the only solution to $A\mathbf{v} = \mathbf{0}$ is $\mathbf{v} = \mathbf{0}$, the kernel of the transformation contains only the zero vector, confirming that T is one-to-one.

Example

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

This matrix represents the transformation that we'll apply to vectors in $\mathbb{R}^2$. The transformation maps any vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ to a new vector $T(\mathbf{v}) = A\mathbf{v}$.

Let's try applying the transformation to a couple of vectors and see what happens.

Step 1: Apply the Transformation

Start with two distinct vectors $\mathbf{v}_1$ and $\mathbf{v}_2$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Now, apply the transformation to each:

$$T(\mathbf{v}_1) = A\mathbf{v}_1 = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (0 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$T(\mathbf{v}_2) = A\mathbf{v}_2 = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} (2 \times 3) + (1 \times 1) \\ (0 \times 3) + (3 \times 1) \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

Step 2: Check for Uniqueness

After applying the transformation, we get:

- $T(\mathbf{v}_1) = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$
- $T(\mathbf{v}_2) = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$

Since $T(\mathbf{v}_1) \neq T(\mathbf{v}_2)$, the transformation maps each input vector to a unique output vector, confirming that the transformation is **one-to-one** for these vectors.

Step 3: The General Case

To check whether this transformation is **one-to-one** in general, we look at the determinant of the matrix A . The determinant of a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is given by:

$$\det(A) = ad - bc$$

For our matrix $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, we compute:

$$\det(A) = (2 \times 3) - (1 \times 0) = 6$$

Since the determinant is non-zero ($\det(A) = 6 \neq 0$), the transformation is invertible, meaning it is **one-to-one** for all vectors in $\mathbb{R}^2$.

In this example, the matrix transformation defined by $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ is a **one-to-one linear transformation**. Every vector in the input space is uniquely mapped to a distinct vector in the output space, ensuring that no information is lost or duplicated in the transformation process.

This concept of one-to-one transformations is vital in many areas, including computer graphics (where preserving distinct points is important), physics (where transformations represent changes in systems), and more.

Reflection: $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Dilation: $A = \begin{pmatrix} 1.5 & 0 \\ 0 & 1.5 \end{pmatrix}$

Identity: $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Rotation: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Shear: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$

Applications to Computer Graphics

Introduction

Computer graphics are images displayed or animated on a computer screen.

Applications of computer graphics are widespread and growing rapidly.

Examples:

- Computer Aided Design (CAD) for various engineering processes
- Gaming – PlayStation, Xbox, etc.
- Entertainment industry – special effects in the movies

Applications to Computer Graphics

Introduction (Continued...)

Most interactive computer software for business and industry makes use of computer graphics in the screen displays and for other functions, such as graphical display of data, desktop publishing, and slide production for commercial and educational presentations. Consequently, anyone studying a computer language invariably spends time learning how to use at least two-dimensional (2D) graphics.

Applications to Computer Graphics

Introduction (Continued...)

This section examines some of the basic mathematics used to manipulate and display graphical images such as a wire-frame model of an airplane. Such an image (or picture) consists of a number of points, connecting lines or curves, and information about how to fill in closed regions bounded by the lines and curves. Often, curved lines are approximated by short straight-line segments, and a figure is defined mathematically by a list of points.

Among the simplest 2D graphics symbols are letters used for labels on the screen. Some letters are stored as wire-frame objects; others that have curved portions are stored with additional mathematical formulas for the curves.

Linear algebra plays a fundamental role in computer graphics, as it provides the mathematical framework for representing and manipulating geometric shapes and their transformations.
Summary how linear algebra is applied in computer graphics:

1. Representation of Points and Vectors

- **Points in Space:** In 2D or 3D graphics, points in space are represented as vectors. For example, a point P in 3D space can be represented as a vector:

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

where x , y , and z are the coordinates of the point in the 3D space.

- **Direction and Magnitude:** Vectors are also used to represent directions and magnitudes (such as velocity or force) in a 3D scene.

2. Geometric Transformations

One of the most important applications of linear algebra in computer graphics is in performing **geometric transformations**. These operations involve transforming points or objects (represented as vectors) using matrices. Common transformations include:

- **Translation:** Moving an object from one position to another. In homogeneous coordinates, translation can be represented using a matrix.
- **Scaling:** Adjusting the size of an object by scaling its coordinates. A scaling transformation matrix can scale an object uniformly or non-uniformly along each axis.

$$S = \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{pmatrix}$$

where s_x , s_y , and s_z are the scaling factors along the x, y, and z axes.

- **Rotation:** Rotating an object around an axis in 2D or 3D. In 2D, for example, a rotation matrix can rotate a point around the origin by an angle θ :

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

In 3D, rotation matrices become more complex, involving the axis of rotation as well.

- **Shearing:** A transformation that distorts the shape of an object by shifting points along one axis based on their position along another axis. A shear matrix in 2D might look like this:

$$S = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

where k is the shear factor.

- **Reflection:** A matrix can also be used to reflect objects across a plane. For example, reflecting across the y-axis in 2D would use the matrix:

$$R = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

3. Homogeneous Coordinates

In computer graphics, **homogeneous coordinates** are used to represent points in space. This allows for the convenient application of transformations (like translation, rotation, scaling) using matrix multiplication. A point (x, y, z) in 3D is represented as $(x, y, z, 1)$ in homogeneous coordinates, enabling translation to be represented as a matrix operation.

4. Projections

Projections are essential in rendering 3D scenes onto a 2D display (like a computer screen). Linear algebra is used to project 3D objects onto 2D surfaces by applying **projection matrices**. Common types of projections include:

- **Orthographic Projection:** This type of projection removes depth information and represents 3D objects on a 2D plane, preserving the relative size of objects regardless of distance. The orthographic projection matrix looks like:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- **Perspective Projection:** In this projection, objects farther from the camera appear smaller, creating a sense of depth. This is achieved through a perspective projection matrix, which applies depth cues based on distance from the camera.

5. Cross Product and Dot Product

Linear algebra provides tools like the **dot product** and **cross product**, which are crucial for computing angles, lighting, and surface normals in computer graphics.

- **Dot Product:** The dot product between two vectors $\mathbf{u}$ and $\mathbf{v}$ measures the cosine of the angle between them, which is used in shading models to calculate how much light hits a surface.

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

- **Cross Product:** The cross product produces a vector that is perpendicular to two given vectors. It's used for finding normal vectors to surfaces, which are important for lighting and reflection calculations.

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} u_y v_z - u_z v_y \\ u_z v_x - u_x v_z \\ u_x v_y - u_y v_x \end{pmatrix}$$

6. Lighting and Shading

In **lighting** calculations, linear algebra is used to determine how light interacts with surfaces. This is typically done using the **Phong lighting model** or other lighting models that rely on vector mathematics:

- **Surface normals** are computed using vectors to determine how light reflects off a surface.
- **Shading models** use dot products to determine the intensity of light on a surface based on the angle between the light source and the surface normal.

7. Rasterization and Viewing Pipeline

Linear algebra underpins the **graphics pipeline**, especially the stages related to **rasterization**, where the continuous 3D world is discretized into pixels on the 2D screen. The process of converting 3D object coordinates to screen coordinates involves several matrix transformations:

- **Model matrix:** Transforms objects from their local coordinate system to world space.
- **View matrix:** Transforms world space coordinates to camera or view space.
- **Projection matrix:** Projects 3D view space onto a 2D screen.

8. Interpolation and Animation

Linear algebra helps in **interpolation** between keyframes in animation. For example, **linear interpolation** (LERP) allows smooth transitions between vectors:

$$LERP(A, B, t) = (1 - t)A + tB$$

where A and B are points (or vectors) and t is a parameter that varies between 0 and 1, allowing smooth transitions between the two points.

9. Computer Vision and Image Processing

- **Image transformations:** Scaling, rotating, or warping an image involves matrix operations on the pixel coordinates.
- **Edge detection and filtering:** Linear algebra techniques, such as **convolution**, are used to apply filters for edge detection, blurring, and sharpening images.

Linear algebra is indispensable in computer graphics as it provides the tools and techniques for transforming and manipulating objects, handling geometry, performing lighting calculations, and managing the entire graphics pipeline. By using vectors, matrices, and operations like dot and cross products, computer graphics engines can create and display complex visual scenes efficiently and realistically.

Applications to Computer Graphics

EXAMPLE 1 The capital letter N in Fig. 1 is determined by eight points, or *vertices*. The coordinates of the points can be stored in a data matrix, D .

$$\begin{array}{l} \text{Vertex:} \\ \begin{array}{c} x\text{-coordinate} \\ y\text{-coordinate} \end{array} \begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \left[\begin{array}{cccccccc} 0 & .5 & .5 & 6 & 6 & 5.5 & 5.5 & 0 \\ 0 & 0 & 6.42 & 0 & 8 & 8 & 1.58 & 8 \end{array} \right] = D \end{array} \end{array}$$

In addition to D , it is necessary to specify which vertices are connected by lines, but we omit this detail. ■

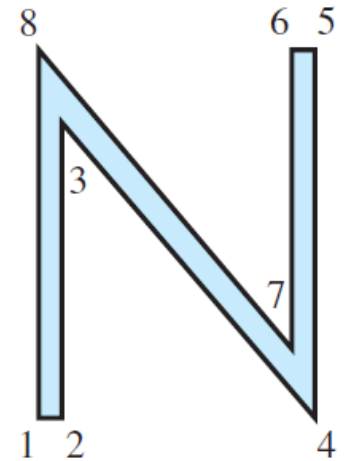
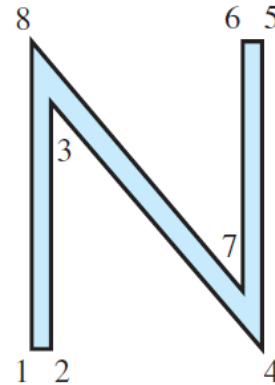
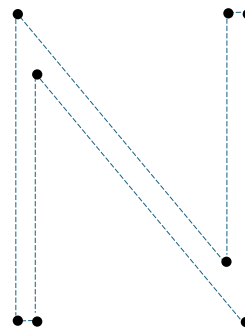


FIGURE 1
Regular N .

Applications to Computer Graphics

The main reason graphical objects are described by collections of straight-line segments is that the standard transformations in computer graphics map line segments onto other line segments.

Once the vertices that describe an object have been transformed, their images can be connected with the appropriate straight lines to produce the complete image of the original object.



Applications to Computer Graphics

EXAMPLE 2 Given $A = \begin{bmatrix} 1 & .25 \\ 0 & 1 \end{bmatrix}$, describe the effect of the shear transformation $\mathbf{x} \mapsto A\mathbf{x}$ on the letter N in Example 1.

SOLUTION By definition of matrix multiplication, the columns of the product AD contain the images of the vertices of the letter N.

$$AD = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 0 & .5 & 2.105 & 6 & 8 & 7.5 & 5.895 & 2 \\ 0 & 0 & 6.420 & 0 & 8 & 8 & 1.580 & 8 \end{bmatrix}$$

The transformed vertices are plotted in Fig. 2, along with connecting line segments that correspond to those in the original figure.

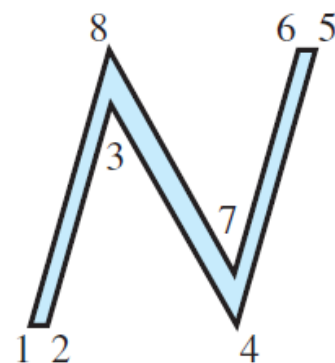


FIGURE 2
Slanted N .